

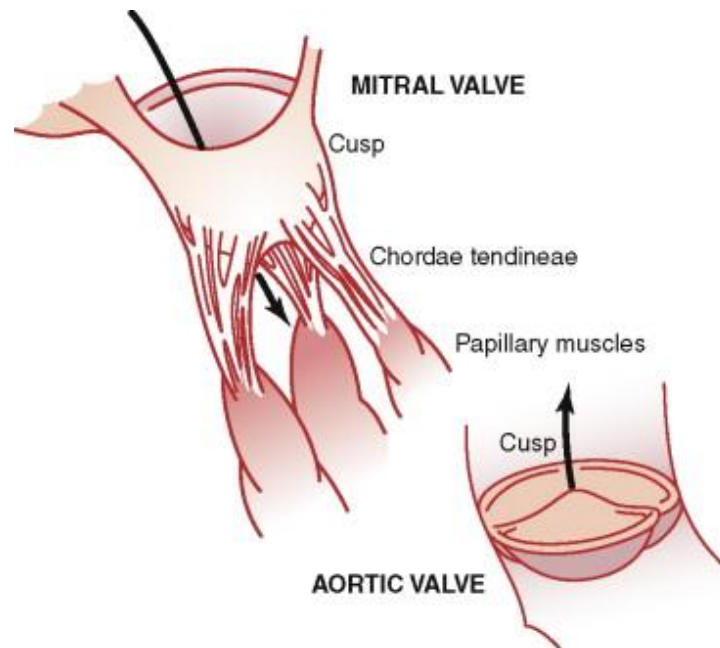
Session Objectives

- 1. Explain pressure/volume changes during each phase of atrial and ventricular systoles and diastoles in a normal cardiac cycle**
- 2. Compare states of heart valves to corresponding phases of cardiac cycle**
- 3. Deconstruct the relationship of electrocardiogram with respect to the events in a cardiac cycle**
- 4. Recognize heart sounds corresponding to a cardiac cycle**
- 5. Map various events of a cardiac cycle in an integrative fashion**

Case study

- A 73-year old man visits ER with complaints of exertional dyspnea, syncope, chest pain. Poorly managed hypercholesterolemia with 11-year history of hypertension.
 - BP: 101/75.
 - Lungs clear. Systolic murmur loudest over the aorta; peaks at mid systole.
 - ECG: LAD, ST normal.
 - CXR: large LV, calcification of aortic valve.
 - Echo: Greatly increased velocity of flow during the systolic portion of the cardiac cycle. Left heart enlarged and LVH.
 - Cardiac enzymes normal.

Valves

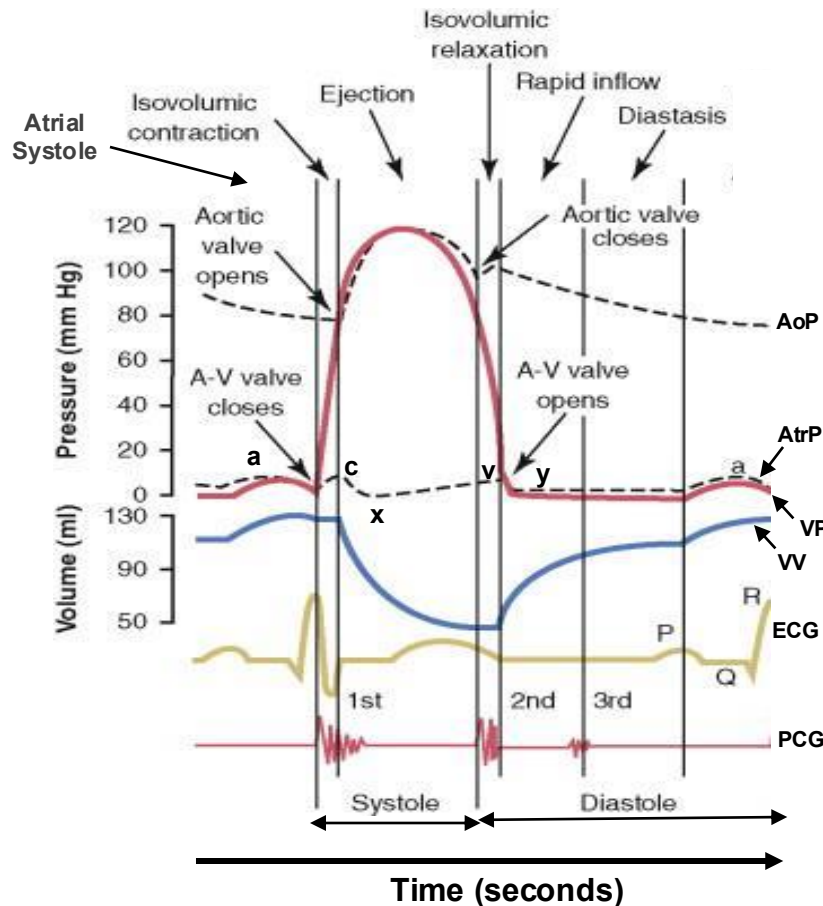


- Beginning of the ventricular systole – the AV valves (e.g., mitral) close
- *Valves are pulled inward toward the ventricles by papillary muscles to prevent their bulging too far backward toward the atria during ventricular contraction*

Background – Cardiac Cycle

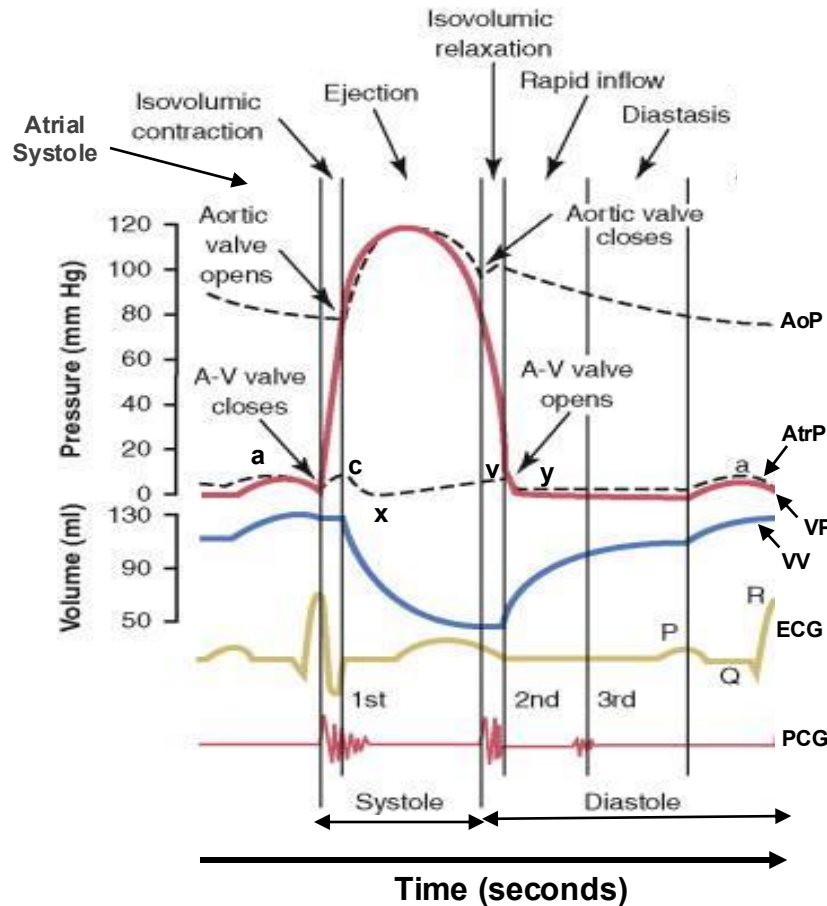
- Atrial (LA and RA) contraction **precedes** ventricular (LV and RV) contraction
 - Delay in signal transmission at the AV node >0.1 s – helps pump blood into the ventricles before the strong ventricular contraction begins.
- Systole and Diastole – different timings in Atria and Ventricles
- AT REST: **2/3** of the cardiac cycle is spent in **diastole**; 1/3 in systole
- INCREASING ACTIVITY LEVELS:
 - less time in diastole
 - **more time in systole**

Atrial pressure waves



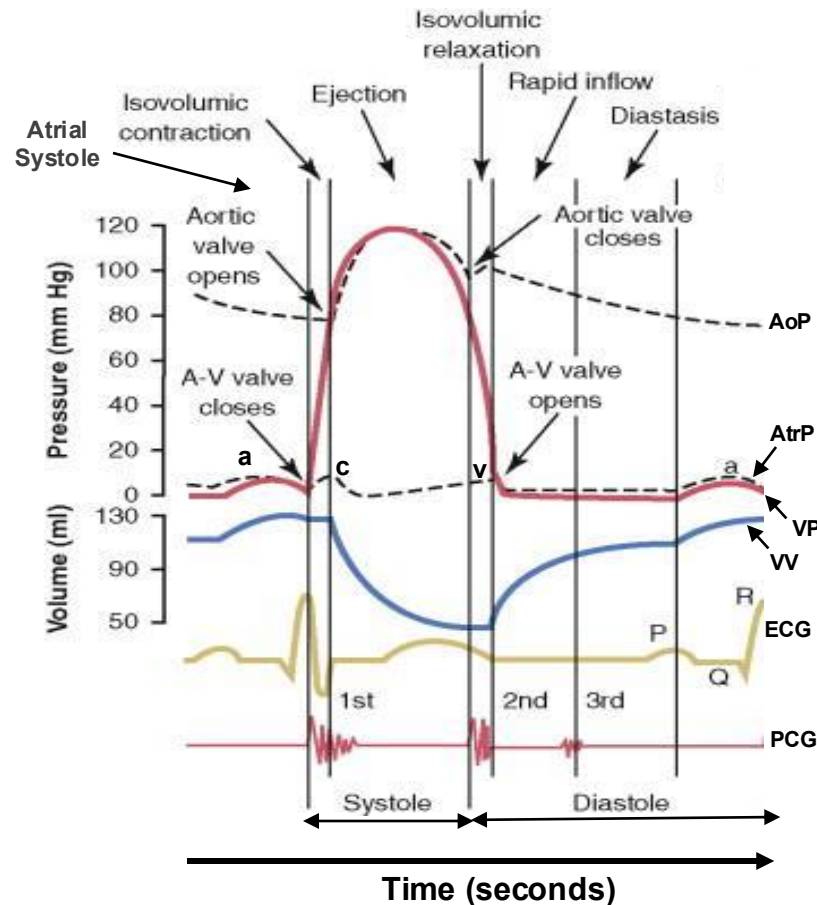
- **"a" wave (a. systole)**
 - small increase in atrial pressure from **atrial contraction**
- **"c" wave (a. diastole)**
 - just after the peak of the QRS complex
 - results from a **slight increase in atrial pressure** due to **atrial compression** during **ventricular contraction**
- **"x" descent**
 - Atrial relaxation

Atrial pressure waves (Contd...)



- **"v" wave (a. diastole)**
 - AV valves closed at the beginning of **ventricular systole**
 - Atrial pressure slowly increases as the **atria fill with incoming blood creating the "v" wave.**
- **"y" descent**
 - Ventricular filling
- AV valves open at beginning of ventricular filling
 - Atrial blood flows into the ventricles
 - Atrial pressure drops

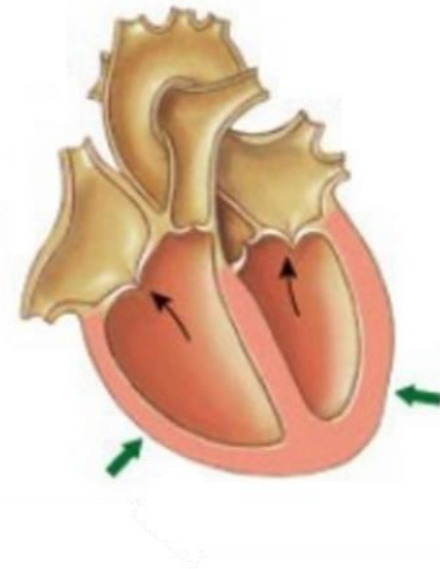
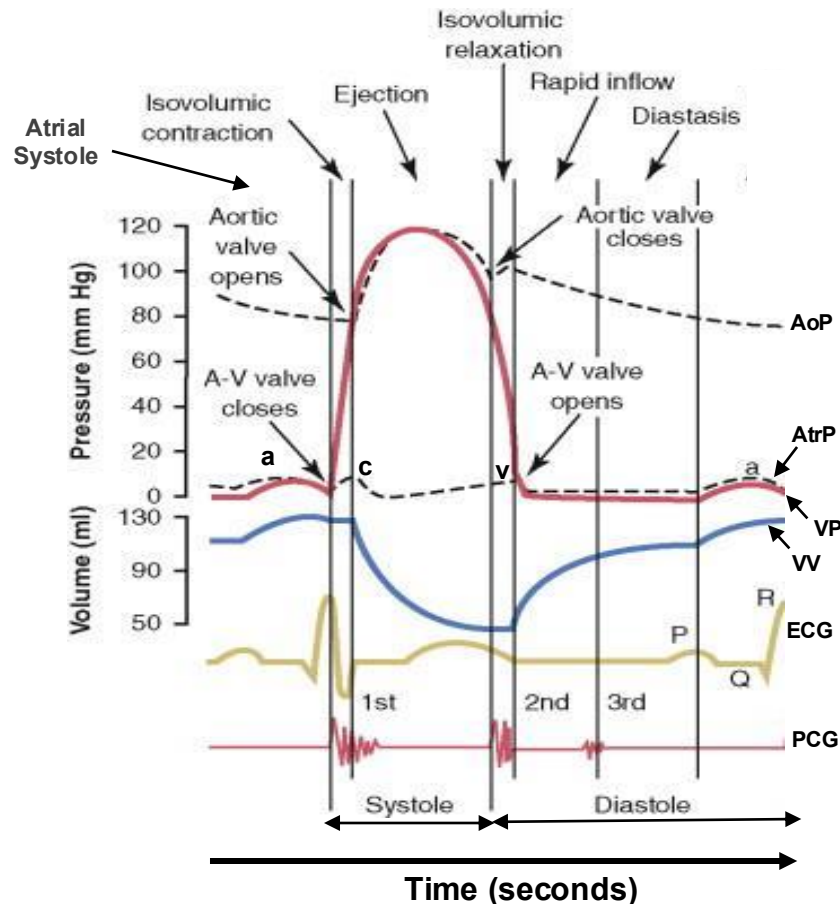
Ventricular systole



- Between start of the QRS complex and (beginning of the) end of the T wave.
- Contraction follows depolarization.
- Ventricular systole occurs in **three phases**
- **End diastolic volume:** Volume of blood in the ventricles at the **start of ventricular contraction (systole)**.
- Ventricular stretch secondary to filling
 - **Frank-Starling law** of the heart (discussed later)

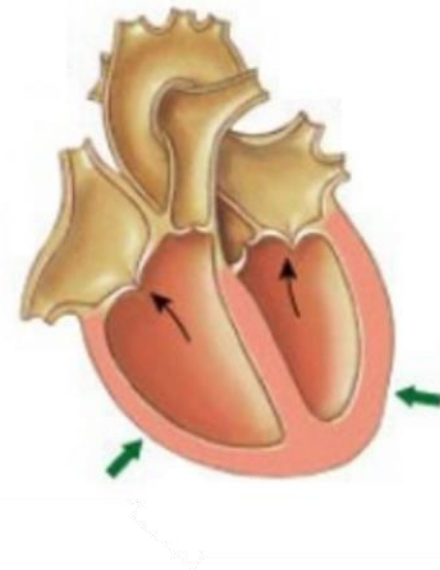
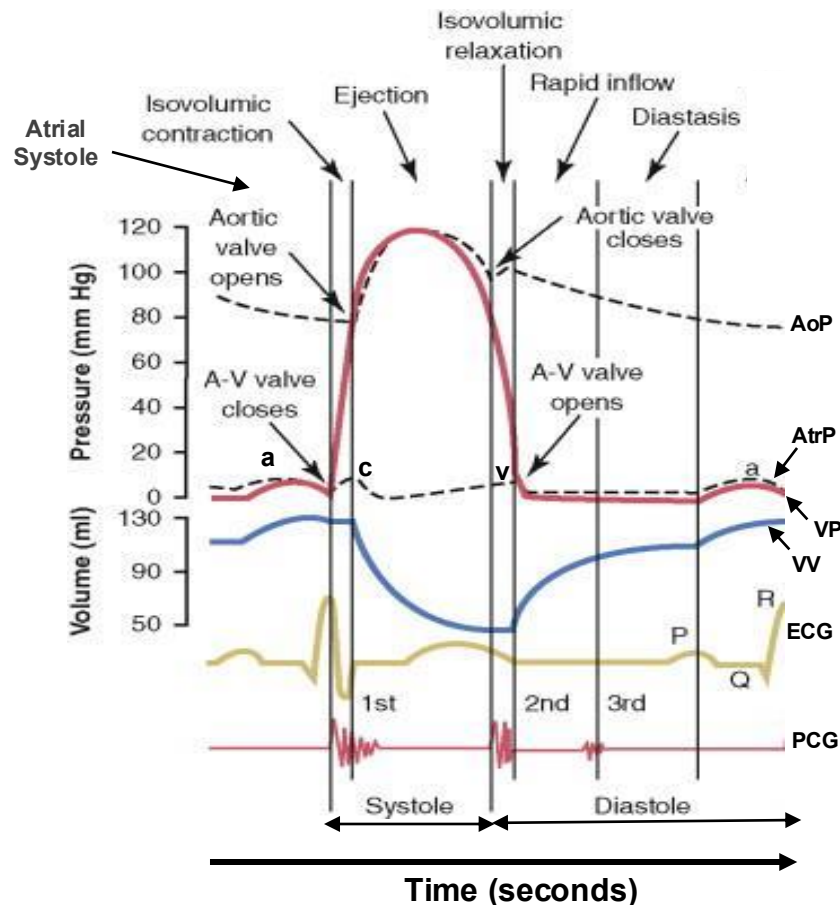
Ventricular systole: Isovolumetric Contraction Phase

- Papillary muscle – initial contraction
- Both valves closed
- Both ventricles are closed, blood-filled chambers
- Since the blood cannot be compressed, the initial effect of ventricular contraction is an **increase in intraventricular pressure without a change in ventricular volume**.



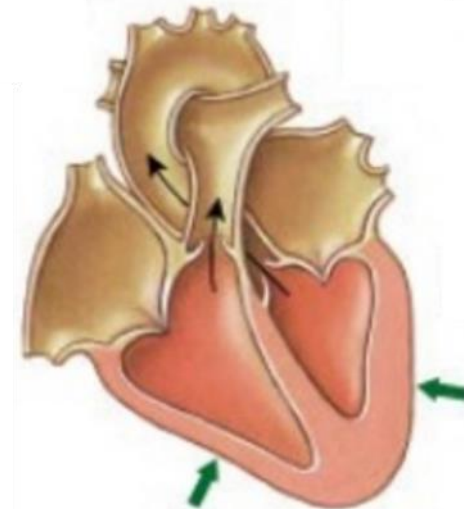
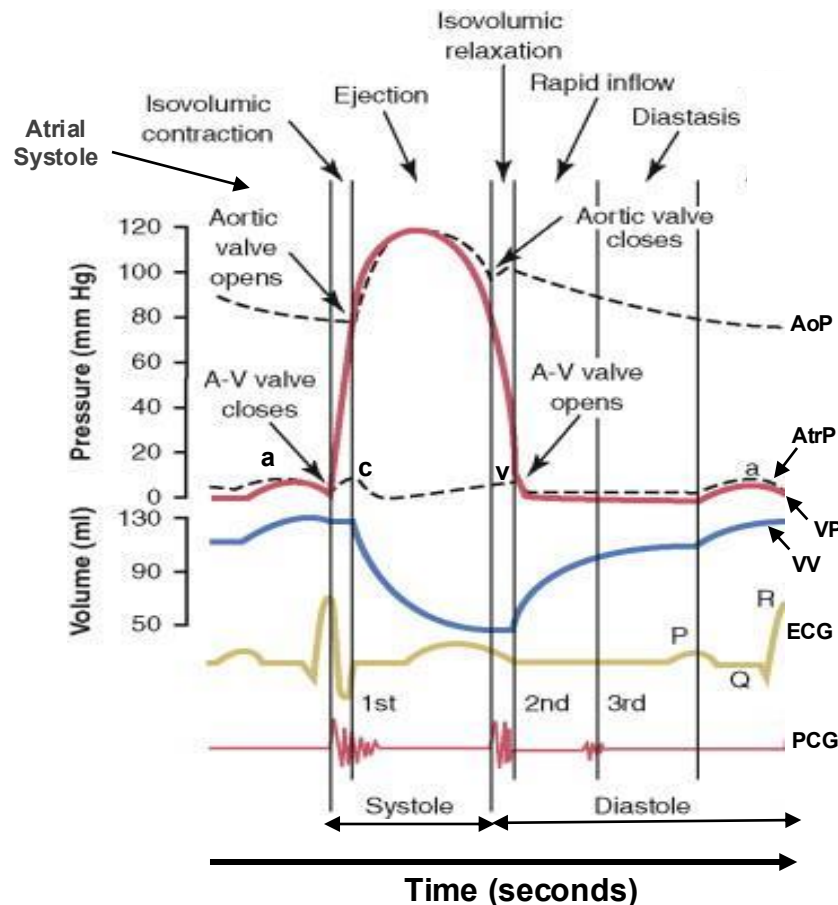
Ventricular systole: Isovolumetric Contraction Phase

- Contraction is isometric
- Both the ventricles develop enough pressure (force) to overcome their respective afterloads (e.g. aortic and PA pressures, discussed later).
- dP/dt (max)



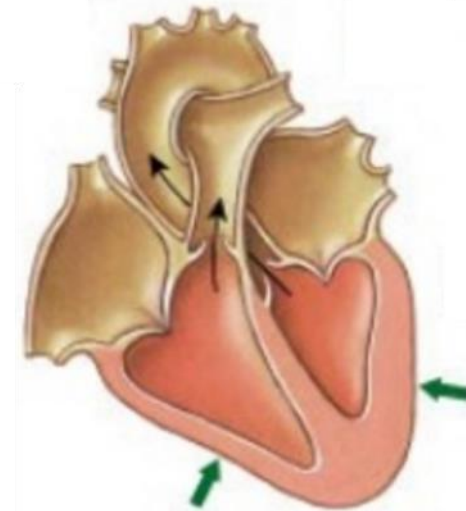
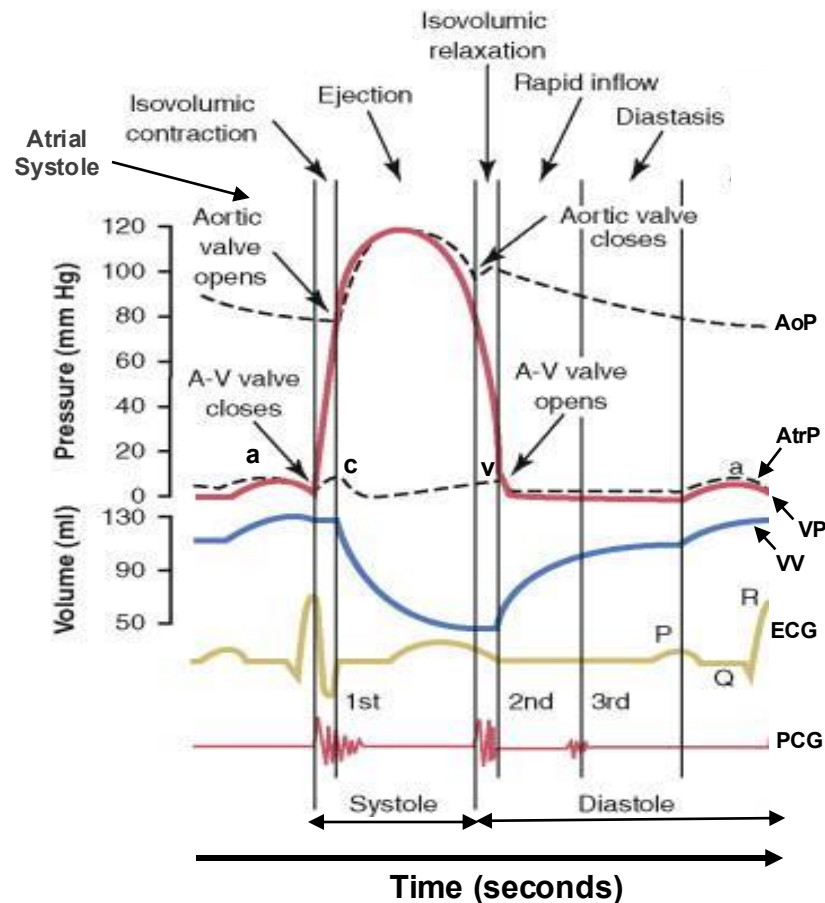
Ventricular systole: Rapid Ejection Phase

- When the **rising ventricular pressure exceeds arterial pressure**.
- The **slightly higher ventricular pressures** force open both semilunar valves and drive blood into the AO or the PA.
- **Ventricular pressure continues to increase** after the AoV/PV have opened because the ventricles are still contracting.



Ventricular systole: Rapid Ejection Phase

- Isotonic contraction
- The aortic (and PA) pressures increase because blood is being forced into them.
- LVP > BP



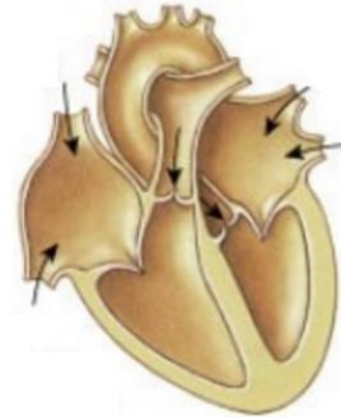
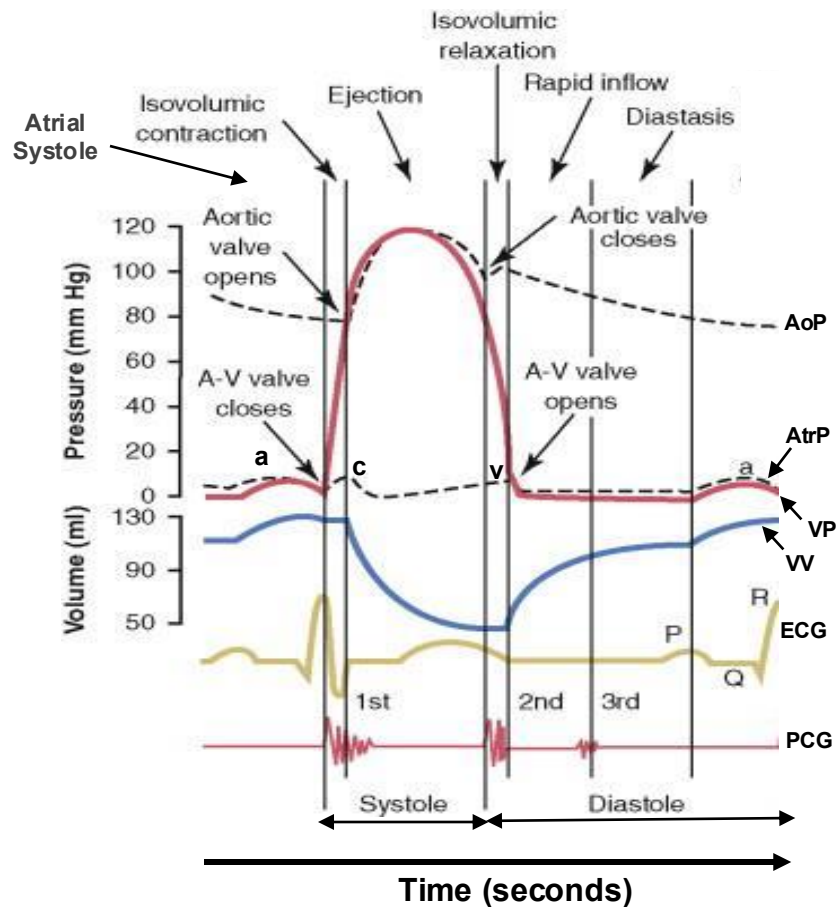
Sample question 1

The rapid ventricular ejection phase changes to the reduced ejection phase when?

- A. Aortic pressure reaches its minimum diastolic value.
- B. Left semilunar valve closes
- C. Left AV valve opens
- D. Pressure in the left ventricle falls below the pressure in the aorta.
- E. Atrial pressure becomes greater than ventricular pressure.

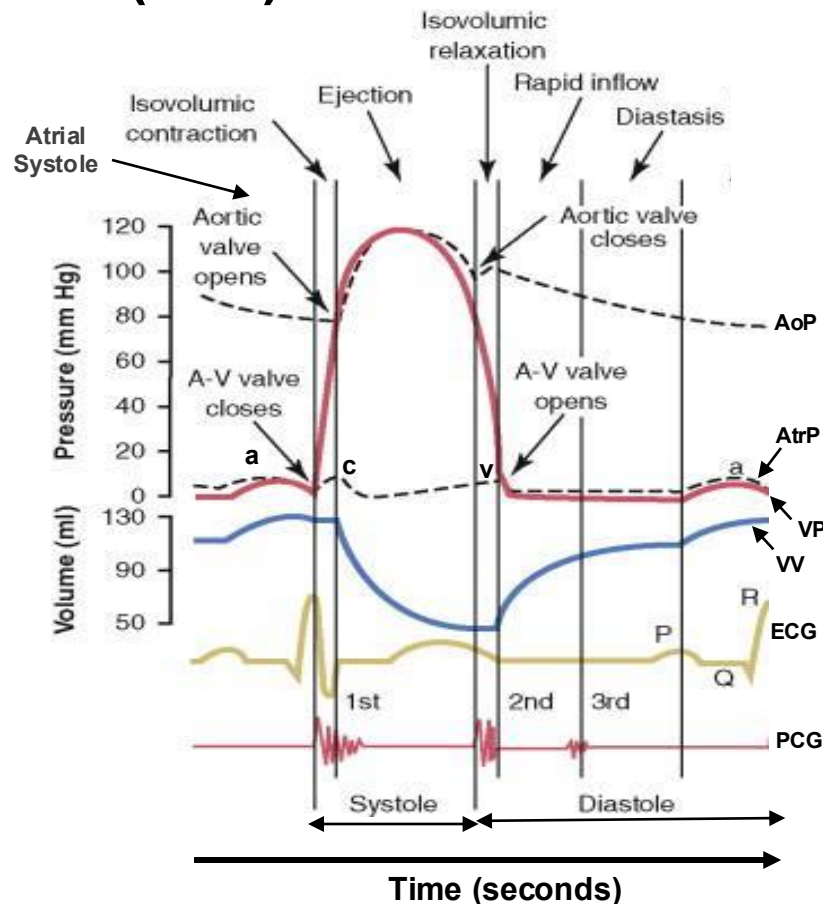
Ventricular diastole: Isovolumetric Relaxation Phase

- Ventricular diastole (VD) – three phases:
- VD starts with the **closure of the semilunar valves** (AV valve also closed)
- Isovolumetric relaxation**



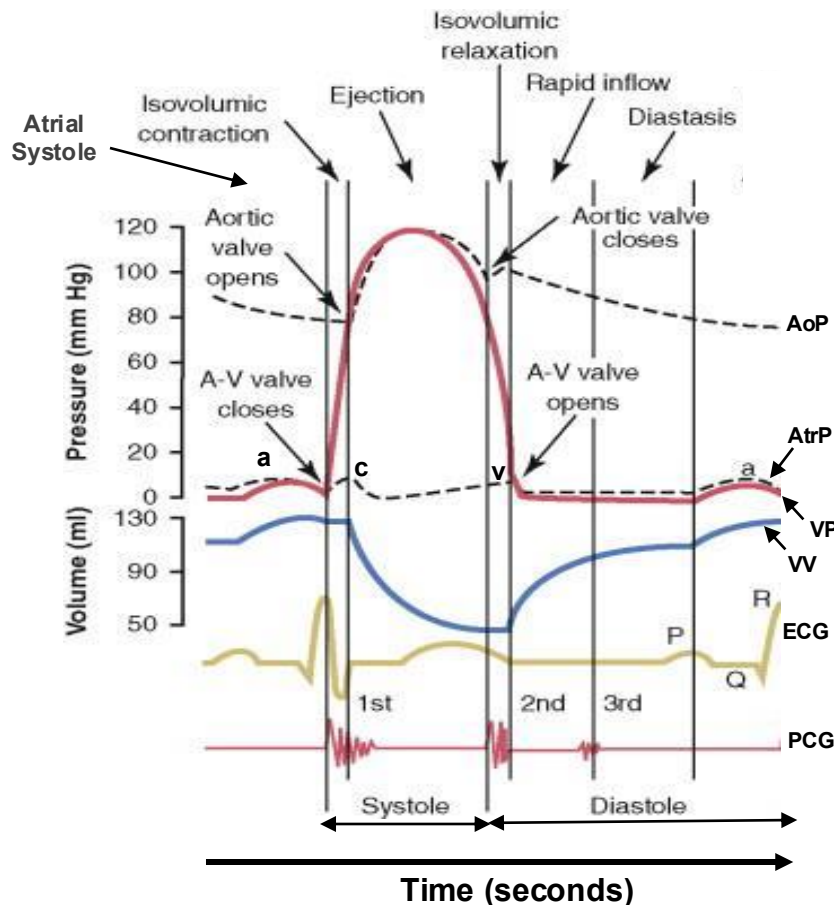
Ventricular diastole: Isovolumetric Relaxation Phase

- Fall in ventricular pressure follows T wave repolarization.
- LV does not completely empty during ejection phase.
- The **volume** of blood remaining in the ventricles at the **end of systole (or start of diastole)** is called end systolic volume (ESV).
- **dP/dt (min)**



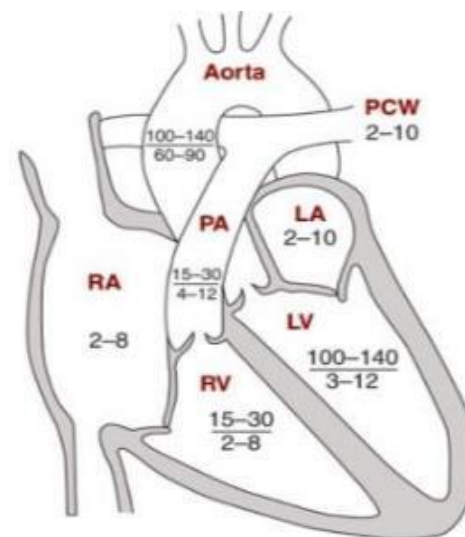
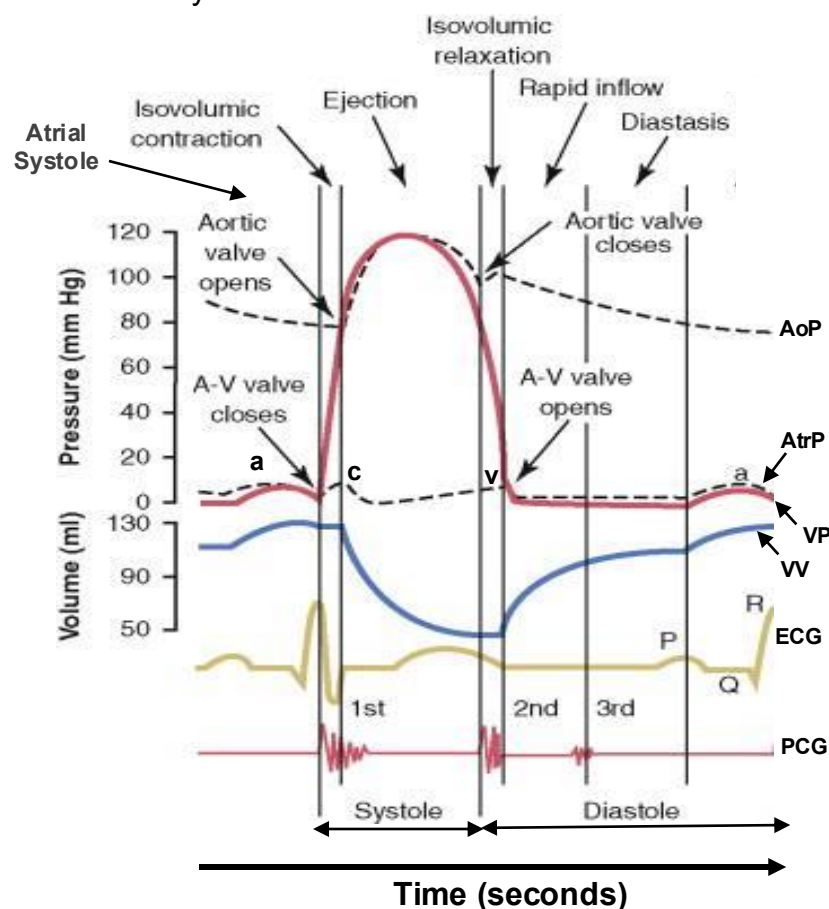
Ventricular diastole: Reduced filling phase

- Blood flow: Veins → relaxed atria → relaxed ventricles
- VD Pressure << Atrial pressure, VD transitions into a slower **reduced filling phase (or diastasis)**.
- The **Diastasis** ends with the **P wave** of the next cycle.
- VD ends with the appearance of R-wave



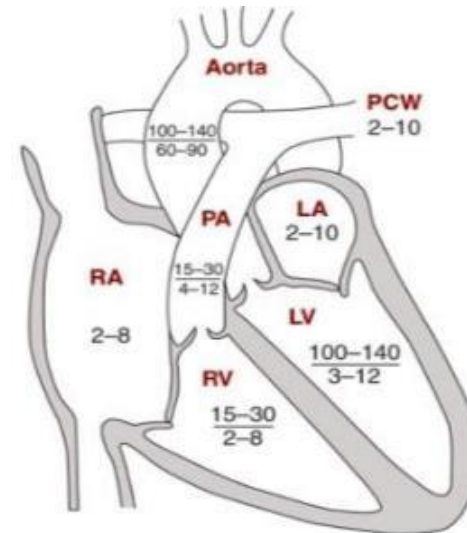
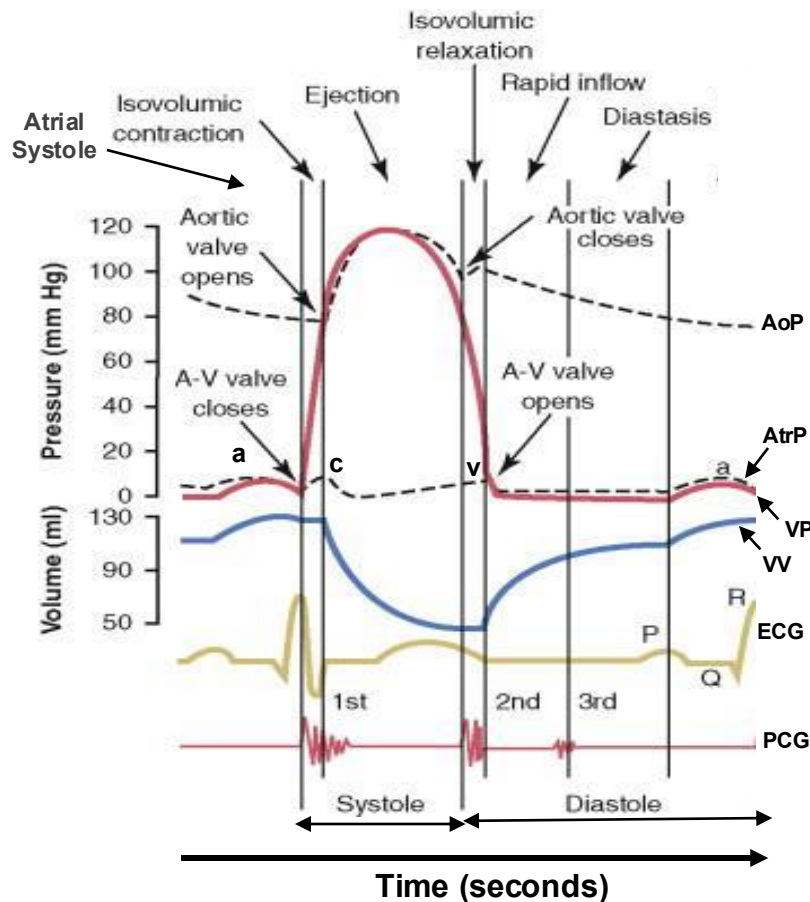
Aortic pressure changes and Pulse pressure

- Max syst. Ao Pressure ~ 120 mm Hg at rest.
- **Rapid** ejection phase – LVP > Syst. Ao. Pressure
- **Reduced** ejection phase – Systolic Ao. Pressure > LVP
 - Ao pressure decreases as LV pressure decreases **until closure of the semilunar valve.**
 - The **dicrotic notch** or **incisura** is caused by a slight back flow of aortic blood that fills the cusps of the SV as they close.



Aortic pressure changes and Pulse pressure (contd...)

- Ventricular diastole
 - Ao Pressure gradually decreases**, and **reaches** its lowest diastolic value, (~ 80 mmHg), just before opening of the SV in the next cycle.
- Blood flows** through the **arteries in spurts** because of the variation in Ao pressure. **Flow in capillaries and veins is steady.**



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